

# Introduction to Amazon Web Services

AWS Technical Essentials — Module 1 of 7

# 🖥️ What Is Cloud Computing?

**On-demand delivery** of IT resources over the internet with **pay-as-you-go** pricing.

- No physical data centers to manage
- Provision resources in minutes
- Scale up or down on demand
- Pay only for what you consume

# 🎯 Six Advantages of Cloud

1. **Trade CapEx for variable expense**
2. **Massive economies of scale**
3. **Stop guessing capacity**
4. **Increase speed and agility**
5. **Stop running data centers**
6. **Go global in minutes**

# 🌍 AWS Global Infrastructure

## 🌏 Regions

- 33+ geographic locations
- Physically isolated
- Data stays in-Region

## 🏧 Availability Zones

- 105+ AZs globally
- One or more data centers
- Low-latency interconnects

## ⚡ Edge Locations

- 600+ Points of Presence
- CloudFront CDN caching
- Route 53 DNS hosting

# 🛠️ Interacting with AWS

### 💻 Console

- Web-based GUI
- Great for exploration

### ⌨️ CLI

- Command-line scripting
- Automate tasks

### 🔌 SDKs

- Language-specific APIs
- Integrate into apps

# 🔒 Shared Responsibility Model

## AWS Responsibility

*"Security OF the cloud"*

- Physical data centers
- Hardware and networking
- Hypervisor and host OS

## Your Responsibility

*"Security IN the cloud"*

- Data and encryption
- IAM and access control
- OS patches and firewalls

# AWS Identity and Access Management

**IAM** controls *who* can access *what* in your AWS account.

- Global service (not Region-specific)
- Free to use, no additional charges
- Principle of least privilege
- Supports MFA for extra security

# 👥 Users, Roles, and Policies

### 👤 Users

- Represent a person or app
- Long-term credentials
- Console or programmatic

### 🎭 Roles

- Temporary credentials
- Assumed by services or users
- Preferred for EC2 apps

### 📜 Policies

- JSON permission documents
- Attach to users or roles
- Allow or deny actions

# 🛠️ Course Application

### Employee Directory Application

Throughout this course, we build a web app layer by layer using AWS services.

- Module 2: EC2 for compute
- Module 3: VPC for networking
- Module 4: S3 for photo storage
- Module 5: DynamoDB for data

# 🛠️ Lab Time

## Introduction to AWS IAM

⏱️ 20 minutes

- Explore IAM users, groups, and roles
- Attach policies and test permissions
- Understand the effect of least privilege